CLASS 19.03.2026

1) Tangent bifurcation
2) Intro transcritical bifurcation

Prop 1

We consider dyn DS $\dot{x} = f_\mu(x)$, $x \in D \subset \mathbb{R}$, $\mu \in \mathbb{R}$
We assume $\exists\, \mu_c, \bar{x}$ s.t. (f regular enough wrt x, μ)

$$f_{\mu_c}(\bar{x}) = f'_{\mu_c}(\bar{x}) = 0$$

and

$$f''_{\mu_c}(\bar{x}) > 0, \quad \partial_\mu f_\mu(\bar{x})\big|_{\mu=\mu_c} < 0.$$

(in general $f''_{\mu_c}(\bar{x})\,\partial_\mu f_\mu(\bar{x})\big|_{\mu_c=\mu} < 0$)

Then,
for $\mu \in B_{\mu_c}$ $\exists$ tangent bifurcation s.t.

(i) $\nexists$ steady state for $\mu < \mu_c$

(ii) $\exists\, \bar{x}_1(\mu), \bar{x}_2(\mu) \in \mathbb{R}$ steady states for $\mu > \mu_c$ s.t.

$$\bar{x}_1(\mu) < \bar{x} < \bar{x}_2(\mu)$$

$\bar{x}_1$: stable, $\bar{x}_2$: unstable

with $|\bar{x}_i(\mu) - \bar{x}| = \mathcal{O}\left(\sqrt{\mu - \mu_c}\right)$.

Wlog $\mu_c = \bar{x} = 0$

Linearization of $f_\mu(x) = f(\mu, x)$ around $(0,0)$ by Taylor,

$$f(\mu,x) = f(0,0) + \partial_x f(0,0)\underbrace{(x)}_{(x-\bar{x})} + \partial_\mu f(0,0)\cdot\underbrace{\mu}_{(\mu-\mu_c)} + \\ + \frac{1}{2}\partial_x^2 f(0,0)\cdot x^2 + R(\mu,x)$$

$$= -\alpha\mu + \beta x^2 + R(\mu,x), \quad \alpha = -\partial_\mu f(0,0) > 0, \quad \beta = \frac{1}{2}\partial_x^2 f(0,0) > 0$$

with $R(\mu,x) = a(\mu,x)\cdot\mu^2 + b(\mu,x)\cdot x^3 + c(\mu,x)\cdot\mu x$

Wlog $+\alpha = \beta = 1$

In fact, we can rescale by defining

$$x = \frac{x'}{\beta} \quad \text{(x' ← new)} \qquad \mu = \frac{\mu'}{\alpha\beta}$$

$$\Rightarrow \frac{d}{dt}x = \frac{1}{\beta}\frac{d}{dt}x' \Bigg\} \Rightarrow \frac{d}{dt}x' = -\mu' + (x')^2 + R\left(\frac{\mu'}{\alpha\beta}, \frac{x'}{\beta}\right)\frac{1}{\beta}$$

$$f\left(\frac{\mu'}{\alpha\beta}, \frac{x'}{\beta}\right) = -\frac{\alpha\mu'}{\alpha\beta} + \beta\frac{(x')^2}{\beta^2} + R\left(\frac{\mu'}{\alpha\beta}, \frac{x'}{\beta}\right)$$

↝ by renaming,

$$\boxed{\frac{d}{dt}x = -\mu + x^2 + R(\mu,x) = g(\mu,x) + R(\mu,x), \quad g(\mu,x) := -\mu + x^2}$$

Rmk Now, you see why we considered the previous example? Thanks to the fact that $\partial_\mu f(0,0) \neq 0$ ⚠

Therefore, * if $\mu < 0$, $|\mu| \ll 1 \Rightarrow \dot{x} > 0 \Rightarrow \nexists$ S.S.

* if $\mu > 0$, $|\mu| \ll 1 \Rightarrow \dot{x} = 0 \Leftrightarrow \bar{x}_{1,2} = \pm\sqrt{\mu}$

Now, I want to rescale the root at $\pm 1 \Rightarrow \underbrace{\bar{x}'_{1,2}}_{\text{new}} = \frac{\bar{x}_{1,2}}{\pm\sqrt{\mu}}$

$\rightsquigarrow \quad x' = \frac{x}{\sqrt{\mu}}$

$$\frac{d}{dt} x' = \frac{1}{\sqrt{\mu}} \frac{d}{dt} x = \frac{1}{\sqrt{\mu}} g\left(\mu, \sqrt{\mu} x'\right) = \frac{1}{\sqrt{\mu}} \left(-\mu + \mu (x')^2 \right) + \frac{1}{\sqrt{\mu}} R\left(\mu, \sqrt{\mu} x'\right)$$

$$= \sqrt{\mu} (x')^2 - \sqrt{\mu} + \frac{1}{\sqrt{\mu}} R\left(\mu, \sqrt{\mu} x'\right)$$

$$= \sqrt{\mu} \underbrace{\left((x')^2 - 1 \right)}_{h(\mu, x')} + \frac{1}{\sqrt{\mu}} R(\quad)$$

Rmk : $h = 0$ at $x' = \pm 1$ (~~even for~~ also with $\mu > 0$).

⚠ The factor in front of $h(\mu, x')$ is essential!
Indeed, $\frac{d}{dt} x' = 0 \iff h(\mu, x') = 0$ (because $\mu > 0$)

However, if $\mu \neq 0 \Rightarrow h = 0$ at $x'_{1,2} = \pm 1$
and the steady states are persistent for $\mu > 0$.
Recalling the rescaling of $f \Rightarrow f$ has 2 zeros of the origin at distance $\mathcal{O}(\sqrt{\mu})$ from the origin that are stable and unstable respectively.

(*) and $\frac{1}{\sqrt{\mu}} R\left(\mu, \sqrt{\mu} x'\right) = \frac{1}{\sqrt{\mu}} \left(\mu^2 + \mu \cdot \sqrt{\mu} x' + \mu^{3/2} (x')^3 \right)$

$$= \mu^{3/2} + \mu x' + \mu^{1} (x')^3$$

$$\Rightarrow \boxed{\frac{d}{dt} x' = \sqrt{\mu} h\left(\mu, x'\right) + \mathcal{O}\left(\mu^{3/2}\right)}$$

i.e. $R\left(\mu, \sqrt{\mu} x'\right) = \mu^{\frac{3}{2}} r(\mu, x')$

(*) with r regular in x, μ.

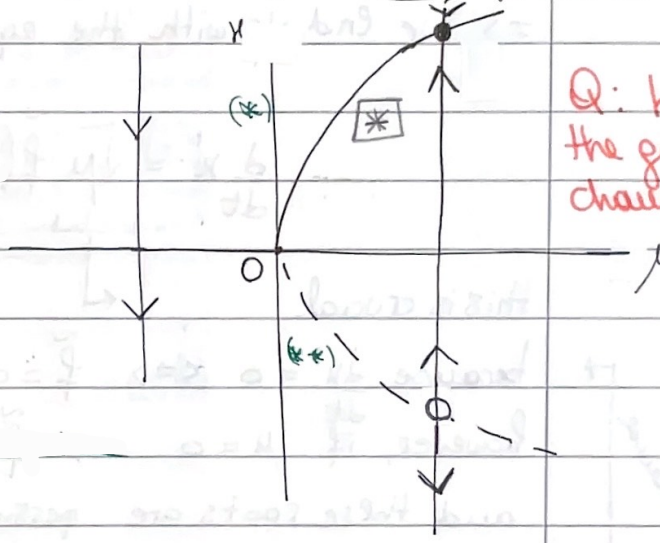
We have already studied an example where by varying the parameter values, a tangent bifurcation occurs.
In the most simple state equations (called normal forms) the equation giving rise to a tangent bif is.

$$\textcircled{*} \quad \boxed{\dot{x}(t) = \mu - x^2(t)} \qquad \mu \in \mathbb{R}$$

→ this normal form satisfies the theorem.

Rmk: it is also called saddle-node bif because in 2D it represents the collision of 2 equi, one saddle point and one node. (Maybe we'll study it in the future).

The representing diagram of $\textcircled{*}$ is



in the course

Q: how does the graph change?

↓ coherent with the Theorem

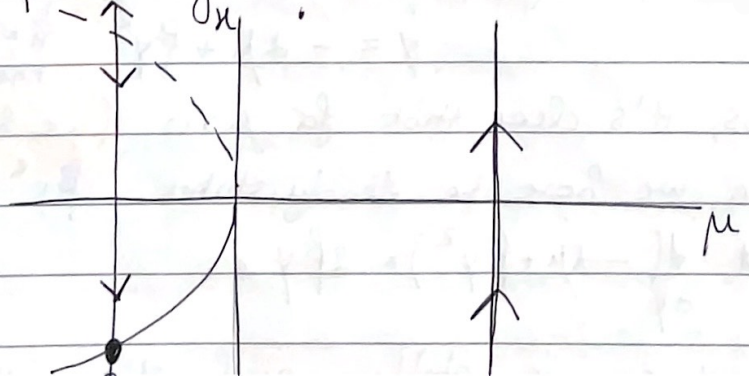
$\bar{x}_1(\mu) < \bar{x} < \bar{x}_2(\mu)$

stable ↙ ↘ unstable

Rmk: If you change the sign of the term of x^2. You end up with ($\mu \in \mathbb{R}$) (still a normal form)

$$\boxed{\dot{x}(t) = \mu + x^2(t)}$$

How the graph changes?



B: Transcritical bifurcation or exchange of stability.

The normal form is:

$$\dot{x}(t) = \mu x(t) + x^2(t) \qquad \mu \in \mathbb{R}$$

$\mu_c = 0$

⚠ $\partial_\mu f|_{\mu_c}(\bar{x}) = \bar{x}_c = 0 \Rightarrow$ No tang. bif.

Then, you have the critical point $\bar{x} = (\pm)\mu$ that is going (↳ ONE NON TRIVIAL) through the origin as μ changes its sign. Therefore, you have a change of stability. Indeed

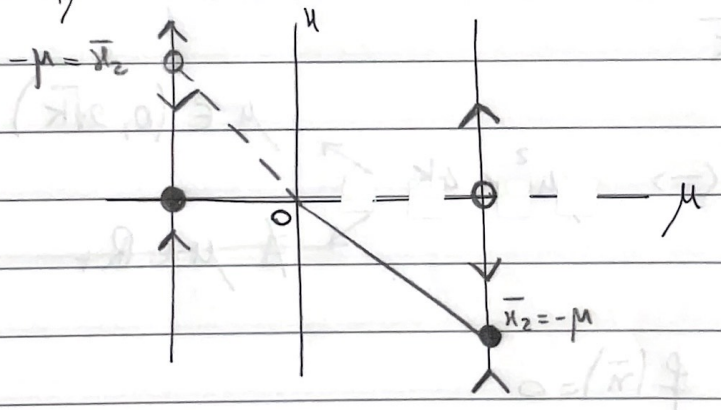
Q: difference with tangent bif?

$\bar{x}_1 = 0$, $\bar{x}_2 = -\mu$, $\mu_c = 0$ and $f_{\mu_c=0}(0) = 0$

$f'_\mu(x) = \mu + 2x \Rightarrow f'_\mu(\bar{x}_1) = f'_\mu(0) = \mu \Rightarrow$ stable, $\mu < 0$; unst, $\mu > 0$

$f'_\mu(\bar{x}_2) = f'_\mu(-\mu) = -\mu \Rightarrow$ stable $\mu > 0$; unst $\mu < 0$

$\dot{x}(t) = x(t)(\mu + x(t))$



Rmk: by changing the sign of quadratic ~~and~~ ~~cubic~~ terms

$$\dot{x}(t) = \mu x(t) - x^2(t), \qquad \mu \in \mathbb{R}$$

Q: how does the graph change?

